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DeepOIS: Gyroscope-Guided Deep Optical Image Stabilizer Compensation

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***Abstract*—Mobile captured images can be aligned using their gyroscope sensors. Optical image stabilizer (OIS) terminates this possibility by adjusting the images during the capturing. In this work, we propose a deep network that compensates for the motions caused by the OIS, such that the gyroscopes can be used for image alignment on the OIS cameras. To achieve this, we first record both videos and gyroscope readings with an OIS camera as training data. Then, we convert gyroscope readings into motion fields. Second, we propose an Essential Mixtures motion model for rolling shutter cameras, where an array of rotations within a frame are extracted as the ground-truth guidance. Third, we train a convolutional neural network with gyroscope motions as input to compensate for the OIS motion. Once finished, the compensation network can be applied for other scenes, where the image alignment is purely based on gyroscopes with no need for images contents, delivering strong robustness. Experiments show that our results are comparable with that of non-OIS cameras, and outperform image-based alignment results with a relatively large margin. Code and dataset is available at: https://github.com/lhaippp/DeepOIS.**

***Index Terms*—Image alignment, gyroscope, optical image stabilizer.**

I. Introduction

IMAGE alignment is a fundamental research problem that has been studied for decades, which has been applied in various applications [1]–[5]. Commonly adopted registration methods include homography [6], [7], mesh-based deformation [2], [8], and optical flow [9]–[12]. These methods look at the image contents for the registration, which usually require rich textures [13], [14] and similar illumination variations [15] for good results, including but not limited to, panorama [1], image/video stitching [2], [3], multi-frame image restoration [4], medical image registration [16], and video stabilization [5]. These methods perform well when the corresponding conditions are satisfied. For example, the performance of homography and mesh-based registration depends on the quality of image features [17], which cannot be guaranteed in scenes with poor textures [13], [14]. Similarly, optical flow may suffer from poor textures, large illumination variations, and large baselines [15]. In summary, in challenging scenarios, such as low-light, low-texture, and moving-foreground, methods based on image contents prone to be erroneous because these methods require the image contents for the registration.

In contrast, gyroscopes can be used to align images, where image contents are no longer required [20]. The gyroscope in a mobile phone provides the camera 3D rotations, which can be converted into homography given camera intrinsic parameters for the image alignment [20], [21]. In this way, the rotational

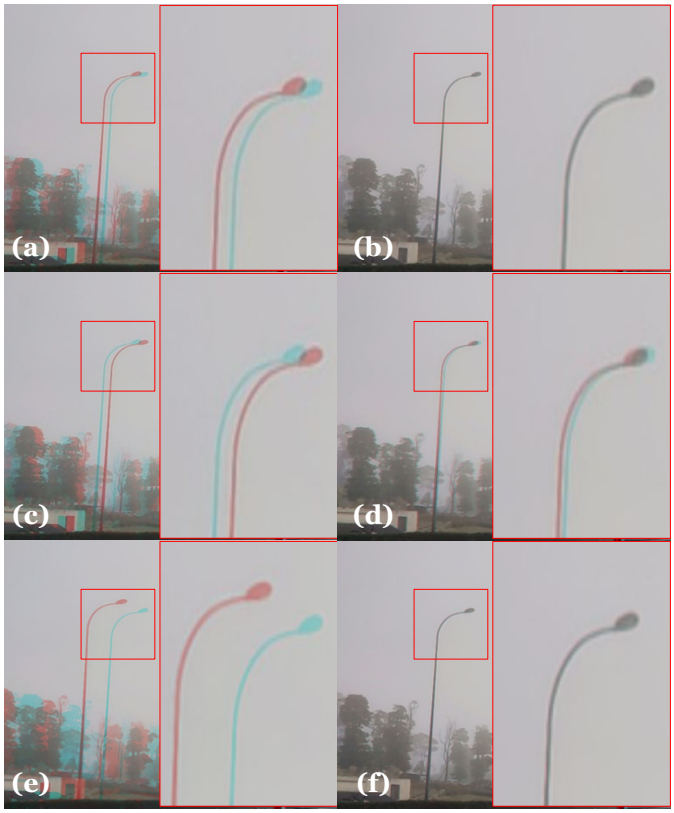


Fig. 1. (a) inputs without the alignment, (b) gyroscope alignment on a non-OIS camera, (c) gyroscope alignment on an OIS camera, (d) deep homography [14] alignment on an OIS camera, (e) SIFT [18]+RANSAC [19] alignment on an OIS camera, and (f) our method on an OIS camera. To demonstrate the results of alignment, we highlight certain areas with the red box which are zoomed on the right side to show more detailed results. We replace the red channel of one image with that of the other image, where misaligned pixels are visualized as colored ghosts. The same visualization is applied for the rest of the paper.

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motions can be compensated. We refer to this as gyro image alignment. One drawback is that translations cannot be handled by the gyro. Fortunately, rotational motions are prominent compared with translational motions [22], especially when filming scenes or objects are not close to the camera [23].

Compared with image-based methods, gyro-based methods are attractive [24]. Firstly, gyro-based methods are irrelevant to image contents, which largely improve the robustness in challenging scenarios where not enough feature points could be detected. In such scenes, gyro-based methods perform much better than image-based methods. Secondly, gyroscopes are widely available and can be easily accessed on daily mobile phones. Many methods have built their applications based on the gyros [25]–[27].

On the other hand, the cameras of smartphones keep evolving, where optical image stabilizer (OIS) becomes more and more popular, which promises less blurry images and smoother videos. It compensates for the 2D pan and tilt motions of the imaging device through lens mechanics [28], [29]. However, OIS terminates the possibility of image registration by gyros. As the homography derived from the gyros no longer corresponds to the captured images, which have been adjusted by OIS with unknown quantities and directions. One may try to read pans and tilts from the camera module. However, this is not easy as it is bounded with the camera sensor, which requires assistance from professionals of the manufacturers [30]. For all we know, Apple does not provide an API to read pans and tilts. Android provides an API named 'OisSample' but invalid for the vast majority of cell phones because it requires the implementation of phone manufacturers. Moreover, even OIS motions can be read by very few phones. They suffer from synchronization problems that could contaminate the data. Even the synchronization problem can be solved, pans and tilts do not directly correspond to the compensated homography.

In this work, we propose a deep learning method that compensates for OIS motions without reading offset data, such that gyroscopes can be used for image alignment on OIS-equipped cell phones. Fig. 1 shows an alignment example in the low-texture environment. To demonstrate the results of alignment, we highlight certain areas with the red box which are zoomed on the right side to show more detailed results. We replace the red channel of one image with the counterpart of the other image, where misaligned pixels are visualized as colored ghosts. Fig. 1 (a) shows two input images. Fig. 1 (b) is the gyro alignment produced by the non-OIS camera. As seen, the images can be well aligned with no OIS interferences. Fig. 1 (c) is the gyro alignment produced by the OIS camera. Misalignments can be observed due to the OIS motion. To further analyze the results of image-based methods in challenging scenes, we compare our method with 2 image-based algorithms. Fig. 1 (d) is the alignment result of deep homography [14]. Fig. 1 (e) is the alignment result of SIFT [18] + RANSAC [19]. Fig. 1 (f) is our OIS compensated result.

Two frames are denoted as I_a and I_b, the motion from gyro between them as G_{ab}. The real motion (after OIS adjustment) between two frames is G'_{ab}. We want to find a mapping function that transforms G_{ab} to G'_{ab}:

$$G'_{ab} = f(G_{ab}). \tag{1}$$

We propose to train a supervised convolutional neural network for this mapping. To achieve this, we record videos and their gyros as training data. The input motion G_{ab} can be obtained directly given the gyro readings. However, obtaining the ground-truth labels for G'_{ab} is non-trivial. We propose to estimate the real motion from the captured images. If we estimate a homography between them, then the translations are included, which is inappropriate for rotation-only gyros. The ground-truth should merely contain rotations between I_a and I_b. Therefore, we estimate an essential matrix and decompose it for the rotation matrix [21]. However, the cell-phone cameras are rolling shutter (RS) cameras, where different rows of pixels have slightly distinct rotation matrices. In this work, we propose an Essential Mixtures model that estimates an array of essential matrices for the RS camera, such that rotational motions can be extracted as the ground-truth. In this way, we can learn the mapping function.

For the evaluation, we collect a testing dataset under various scenes, where we manually mark point correspondences for quantitative metrics. According to our experiments, our network can accurately recover the mapping, achieving gyro alignments comparable to non-OIS cameras. In summary, our contributions are:

- We propose a new problem that compensates OIS motions for gyro image alignment on cell phones. To the best of our knowledge, the problem is not explored yet, but important to many image and video applications.
- We propose a solution that learns the mapping function between gyro motions and real motions, where an Essential Mixtures model is proposed under the RS setting for the real motions.
- We propose a dataset for the evaluation. Experiments show that our method works well compared with non-OIS cameras, and outperforms image-based opponents in challenging cases.

II. Related Work

A. Image Alignments

Homography [6], mesh-based [2], and optical flow [15] methods are the most commonly adopted motion models, which align images in a global, middle, and pixel level. They are often estimated by matching image features or optimize photometric loss [13]. Apart from classical traditional features, such as SIFT [18], SURF [31], and ORB [32], deep features have been proposed for improving robustness, e.g., LIFT [33] and SOSNet [34]. Registration can also be realized by deep learning directly, such as deep homography estimation [14], [35]. In general, without extra sensors, these methods align images based on the image contents.

B. Gyroscopes

Gyroscope is important in helping estimate camera rotations during mobile capturing. The fusion of gyroscope and visual measurements has been widely applied in various applications,

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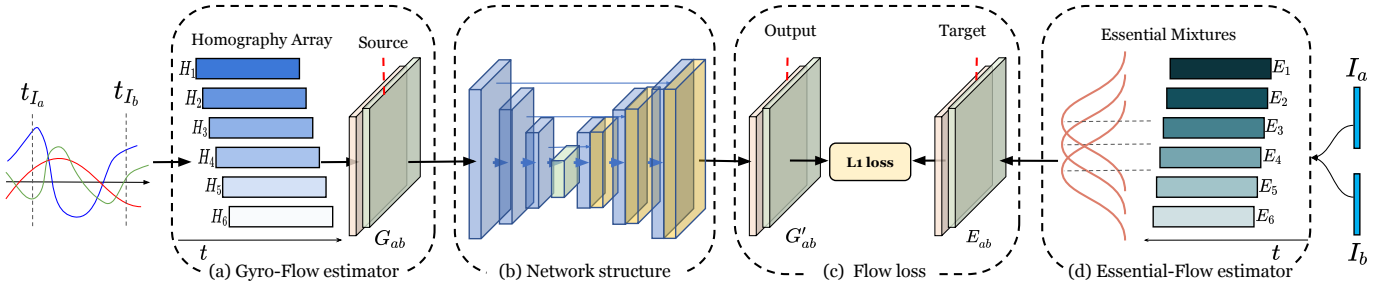


Fig. 2. The overview of our algorithm which includes (a) gyro-based flow estimator, (d) the essential-based flow estimator, and (b) neural network predicting an output flow. For each pair of frames I_a and I_b, the homography array is computed using the gyroscope readings from t_{I_a} to t_{I_b}, which is converted into the source motion G_{ab} as the network input. On the other side, we estimate an Essential Mixtures model to produce the target flow E_{ab} as the guidance. The network is then trained to produce the output G'_{ab}.

including but not limited to image alignment and video stabilization [20], image deblurring [36], simultaneous localization and mapping (SLAM) [25], gesture-based user authentication on mobile devices [26], and human gait recognition [27]. In mobiles, one important issue is the synchronization between the timestamps of gyros and video frames, which requires gyro calibration [37]. In this work, we access the gyro data at the Hardware Abstraction Layer (HAL) of the android layout [38], to achieve accurate synchronizations.

C. Optical Image Stabilizer

Optical Image Stabilizer (OIS) has been around commercially since the mid-90s [39] and becomes more and more popular in our daily cell phones. Since 2016, almost all manufacturers equip their flagship smartphones with OIS. Both the image capturing and video recording can benefit from OIS, producing results with less blur and improved stability [30]. It works by controlling the path of the image through the lens and the image sensor, which is achieved by measuring the camera shakes using high-frequency and precise inertial measurement sensors such as gyroscope. Then it moves the lens horizontally or vertically to counteract shakes by electromagnet motors [28], [29]. While the optical compensation can eliminate motion blur, OIS has some limitations. First, OIS is not suitable for large motion, because it can only compensate for 1-2 degrees moving as it has a limited physical range. Second, if OIS shifts too far from the origin position, some other artifacts are imported to the resulting image such as vignetting or inaccurate focusing. Besides the above disadvantages, one major problem is that OIS cannot be turn-off easily [40], which is not friendly to mobile developers who need gyros to align images. In this work, we enable the gyro image alignment on OIS cameras.

III. Algorithm

Our method is built upon convolutional neural networks. As the control algorithm of OIS only obtains the relative angular displacement of the current time [41], our method takes one gyro-based flow G_{ab} from the source frame I_a to the target frame I_b as input and produces OIS compensated flow G'_{ab} as output. Our pipeline consists of three modules: a gyro-based flow estimator, an Essential Mixtures flow estimator, and a fully convolutional network that compensates for the OIS motion. Fig. 2 illustrates the pipeline. First, the gyro-based flows are generated according to the gyro readings (Fig. 2(a) and Sec. III-A), then they are fed into a network to produce OIS compensated flows G'_{ab} as output (Fig. 2(b) and Sec. III-C). To obtain the ground-truth rotations, we propose an Essential Mixtures model, so as to produce the Essential Mixtures flows E_{ab} (Fig. 2 (d) and Sec. III-B) as the guide to the network (Fig. 2 (c)). During the inference, the Essential Mixtures model is not required. The gyro readings are converted into gyro-based flows and fed to the network for compensation.

A. Gyro-Based Flow

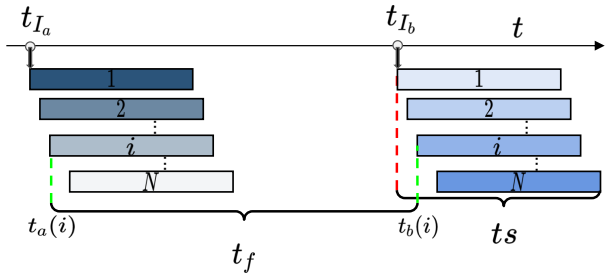


Fig. 3. Illustration of rolling shutter frames. t_{I_a} and t_{I_b} are the frame starting time. t_s is the camera readout time and t_f denotes the frame period ($t_f > t_s$). $t_a(i)$ and $t_b(i)$ represent the starting time of patch i in I_a and I_b.

We compute rotations by compounding gyro readings consisting of angular velocities and timestamps. In particular, we read them from the HAL of android architecture for synchronization. Notably, there are three main layers in the android layout from top to bottom, which are the Framework and Application layer, the External Libraries and Runtime layer, and the HAL layer. We find that the closer to the bottom, the more accurate the gyroscopic data is, avoiding the problem such as synchronization. This is because the time-delay between each layer causes misalignments between the angular velocity readings and the timestamps, e.g., the interference of operating system. We find that this is crucial for the whole system. The rotation vector $n = (\omega_x, \omega_y, \omega_z) \in \mathbb{R}^3$ is computed from gyro readings between frames I_a and

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I_b [20]. The rotation matrix $R(t) \in SO(3)$ can be produced according to the Rodrigues Formula [42].

If the camera is global shutter, the homography is modeled as:

$$H(t) = KR(t)K^{-1}, \tag{2}$$

where K is the intrinsic camera matrix and $R(t)$ denotes the camera rotation from I_a to I_b.

In an RS camera, the 'epipolar lines' becomes 'epipolar curves' [43], as every row of the image is exposed at a slightly different time. Therefore, Eq.(2) is not applicable and a global homography matrix leads to erroneous since every row of the image has slightly different rotation matrices. In practice, assigning each row of pixels with a rotation matrix is unnecessary. We group several consecutive rows into a row patch and assign each patch with a rotation matrix. Fig. 3 shows an example. Let t_s denotes the camera readout time that is the time duration between the exposure of the first row and the last row of pixels:

$$t_a(i) = t_I + t_s \frac{i}{N}, \tag{3}$$

where $t_a(i)$ denotes the start of the exposure of the i-th patch in I_a as shown in Fig. 3, t_I denotes the starting timestamp of the corresponding frame, N denotes the number of patches per frame. The end of the exposure is:

$$t_b(i) = t_a(i) + t_f, \tag{4}$$

where $t_f = 1/FPS$ is the frame period. Here, the homography between the i-th row at frame I_a and I_b can be modeled as:

$$H = KR\left(t_b\right) R^{\top}\left(t_a\right) K^{-1}, \tag{5}$$

where $R\left(t_b\right) R^{\top}\left(t_a\right)$ can be computed by accumulating rotation matrices from t_a to t_b.

In our implementation, we divide the image into 6 patches which computes a homography array containing 6 horizontal homographies between two consecutive frames. We convert the homography array into a flow field [36] so that it can be fed as input to a convolutional neural network. For every pixel p in the I_a, we have:

$$p' = H(t)p, \quad (u, v) = p' - p, \tag{6}$$

computing the offset for every pixel produces a gyro-based flow G_{ab}.

B. Essential Mixtures

Before introducing our model of Essential Mixtures, we briefly review the process of estimating the essential matrix. If the camera is global-shutter, every row of the frame is imaged simultaneously at a time. Let p_1 and p_2 be the projections of the 3D point X in the first and second frame, $p_1 = P_1X$ and $p_2 = P_2X$, where P_1 and P_2 represent the projection matrices. Given the camera intrinsic matrix K. The essential matrix satisfies the equation [21]:

$${p_1}^{\top} K^{-T} E K^{-1} p_2 = 0 \tag{7}$$

where $p_1 = (x_1, y_1, 1)^T$ and $p_2 = (x_1', y_1', 1)^T$.

Then we compute calibrated matching points $\hat{p_1} = K^{-1}p_1$ and $\hat{p_2} = K^{-1}p_2$, then we can rewrite Eq.(7) as:

$${\hat{p_1}}^{\top} E \hat{p_2} = 0, \tag{8}$$

where $\hat{p_1} = (\hat{x_1}, \hat{y_1}, 1)^T$ and $\hat{p_2} = \left(\hat{x_1}', \hat{y_1}', 1\right)^T$. Let f be the 9-element vector made up of E, then Eq.(7) can be written as:

$$\left(\hat{x_1}'\hat{x_1}, \hat{x_1}'\hat{y_1}, \hat{x_1}', \hat{y_1}'\hat{x_1}, \hat{y_1}'\hat{y_1}, \hat{y_1}', \hat{x_1}, \hat{y_1}, 1\right) f = 0, \tag{9}$$

given n correspondences, yields a set of linear equations:

$$Af = \begin{bmatrix} \hat{x_1}'{\hat{p_1}}^T & \hat{y_1}'{\hat{p_1}}^T & {\hat{p_1}}^T \\ \vdots & \vdots & \vdots \\ \hat{x_n}'{\hat{p_n}}^T & \hat{y_n}'{\hat{p_n}}^T & {\hat{p_n}}^T \end{bmatrix} f = 0. \tag{10}$$

Using at least 8 matching points yields a homogenous linear system, which can be solved under the constraint $\|f\|_2 = 1$ using the Singular Value Decomposition(SVD) of $A = UDV^{\top}$ where the last column of V is the solution [21].

In the case of the RS camera, Eq.(7) does not hold anymore, because projection matrices P_1 and P_2 vary across rows instead of being frame-global. Therefore, we introduce Essential Mixtures to assign each row patch with an essential matrix as illustrated in Fig. 4.

To achieve it, we first detect FAST features [44] and track them by KLT [45] between frames. A uniform feature distribution is crucial to model the Essential Mixtures model, so we modify the detection threshold as [3], [46]. Moreover, to model RS effects, we divide a frame into N patches, resulting in N unknown essential matrices E_i to be estimated per frame. If we estimate each essential matrix independently, discontinuity is unavoidable. We propose to smooth neighboring matrices during the estimation as shown in Fig. 2 (d), where a point $\hat{p}_1$ not only contributes to its own patch but also influences its nearby patches weighted by the distance. The essential matrix for point $\hat{p_1}$ is then the mixture:

$$E(\hat{p}_1) = \sum_{i=1}^{N} E_i w_i(\hat{p}_1), \tag{11}$$

where $w_i(\hat{p}_1)$ is the gaussian weight with the mean equaling to the middle of each patch, and the sigma $\sigma = 0.001 * h$, where h represents the frame height.

To fit an Essential Mixtures E_i given a pair of calibrated matching points $(\hat{p}_1, \hat{p}_2)$, we rewrite Eq.(7) as:

$$0 = \hat{p}_1^T E_{\hat{p}_1} \hat{p}_2 = \sum_{i=1}^{N} w_i\left(\hat{p}_1\right) \cdot \hat{p}_1^T E_i \hat{p}_2 \tag{12}$$

where $\hat{p}_1^T E_i \hat{p}_2$ can be transformed into:

$$A^i_{\hat{p}_1} f_i = \begin{pmatrix} \hat{x}_1'\hat{p}_1^T & \hat{y}_1'\hat{p}_1^T & \hat{p}_1^T \end{pmatrix} f_i, \tag{13}$$

where f_i denotes the vector formed by concatenating the columns of E_i. Combining Eq.(12) and Eq.(13) yields a $1 \times 9i$ linear constraint:

$$\underbrace{\left(w_1(\hat{p}_1)A^1_{\hat{p}_1} \ldots w_i(\hat{p}_1)A^i_{\hat{p}_1}\right)}_{A_{\hat{p}_1}} \underbrace{\begin{pmatrix} f_1 \\ \vdots \\ f_i \end{pmatrix}}_{f} = A_{\hat{p}_1} f = 0. \tag{14}$$

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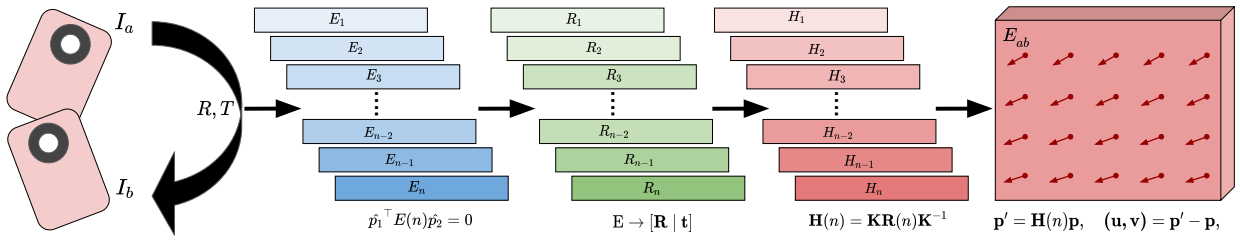


Fig. 4. Illustration of computing Essential Mixtures. Given 2 consecutive frames, an array of essential matrix can be computed, i.e., Essential Mixtures. Given the essential mixtures, rotation matrix and translation matrix can be produced by decomposing essential matrix, so Essential Mixtures is transformed into a set of rotation matrix. Follow the Eq. III-A, an array of homography can be recovered which is then converted into motion field E_{ab}.

Aggregating all linear constraints $A_{\hat{p}_j}$ for every calibrated match point $(\hat{p}_j, \hat{p}_{j+1})$ yields a homogenous linear system $Af = 0$ that can be solved under the constraint $\|f\|_2 = 1$ via SVD.

For robustness, if the number of feature points in one patch is inferior to 8, Eq.(14) is under constrained. Therefore, we add a regularizer to constrain $\lambda \left\| A^i_{\hat{p}} - A^{i-1}_{\hat{p}} \right\|_2 = 0$ to the homogenous system with $\lambda = 1$.

During the estimation, we apply RANSAC [47] in our mixture model to reject outliers. In the first iteration, we select a random sample of 8 matching points per patch to compute the fundamental mixture. Then in each patch, for each putative correspondence we calculate the symmetric epipolar distance [48] $d_\perp$:

$$d_\perp = \left| p_i^\top E_p p_{i+1} \right| \left(\frac{1}{\left\| {E_p}^\top p_i \right\|_2} + \frac{1}{\left\| E_p p_{i+1} \right\|_2} \right), \qquad (15)$$

where E_p denotes the essential matrix in the $p\text{-}th$ patch. We compute the number of inliers where $d_\perp < threshould = 0.2$, and choose the essential mixture with the largest number of inliers.

1) Rotation-Only Homography: Given the essential matrix E_i, it can be decomposed into camera rotations and translations, where only rotations R_i are retained. We use R_i to form a rotation-only homography similar to Eq.(2) and convert the homography array into a flow field as Eq.(6). We call this flow field as Essential Mixtures Flow E_{ab}. Note that, R_i is spatially smooth. E_i is smooth, so does E_{ab}.

C. Network Structure

The architecture of the network is shown in Fig. 2. It utilizes a backbone of UNet [49] which consists of a series of convolutional and downsampling layers with skip connections. The input to the network is gyro-based flow G'_{ab} and the ground-truth target is Essential Mixtures flow E_{ab}. Our network aims to produce a flow of size $2 \times H \times W$ which compensates for motions generated by OIS between G'_{ab} and E_{ab}. Besides, the network is fully convolutional which accepts input of arbitrary sizes. Table I-II shows the network configuration of our model. Given 2 input frames, we first resize frames and adjust the camera intrinsic parameters to compute a gyro-based flow G_{ab} at the spatial resolution of 800x608. Our encoder consists of 5 convolutional layers that downscales the flow map from 2x608x800 to 512x50x38. Each convolutional layer is followed by an Instance Normalization [50] and a ReLU activation function. For the decoder which also contains 5 convolutional layers, it upsamples the feature maps from 512x50x38 back to 2x608x800 with skip connections [51].

TABLE I
NETWORK CONFIGURATION FOR ENCODER.

Layer	Kernel	Strided	Input shape	Output shape
Conv1	3x3	1	2x608x800	32x608x800
Conv2	3x3	1	32x608x800	32x608x800
MaxPool1	2x2	2	32x608x800	32x304x400
Conv3	3x3	1	32x304x400	64x304x400
Conv4	3x3	1	64x304x400	64x304x400
MaxPool2	2x2	2	64x304x400	64x152x200
Conv5	3x3	1	64x152x200	128x152x200
Conv6	3x3	1	128x152x200	128x152x200
MaxPool3	2x2	2	128x152x200	128x76x100
Conv7	3x3	1	128x76x100	256x76x100
Conv8	3x3	1	256x76x100	256x76x100
MaxPool4	2x2	2	256x76x100	256x38x50
Conv9	3x3	1	256x38x50	512x38x50
Conv10	3x3	1	512x38x50	512x38x50

TABLE II
NETWORK CONFIGURATION FOR DECODER. * INDICATES THE FEATURE FROM ENCODER VIA SKIP CONNECTION.

Layer	Kernel	Strided	Input shape	Output shape
UpSample	2x2	-	512x38x50	512x76x100
Concat	-	-	512 + 256*	768
Conv1	3x3	1	768x76x100	256x76x100
Conv2	3x3	1	256x76x100	256x76x100
UpSample	2x2	-	256x76x100	256x152x200
Concat	-	-	256 + 128*	384
Conv3	3x3	1	384x152x200	128x152x200
Conv4	3x3	1	128x152x200	128x152x200
UpSample	2x2	-	128x152x200	128x304x400
Concat	-	-	128 + 64*	192
Conv5	3x3	1	192x304x400	64x304x400
Conv6	3x3	1	64x304x400	64x304x400
UpSample	2x2	-	64x304x400	64x608x800
Concat	-	-	64 + 32*	96
Conv7	3x3	1	96x608x800	32x608x800
Conv8	3x3	1	32x608x800	32x608x800
Conv9	3x3	1	32x608x800	2x608x800

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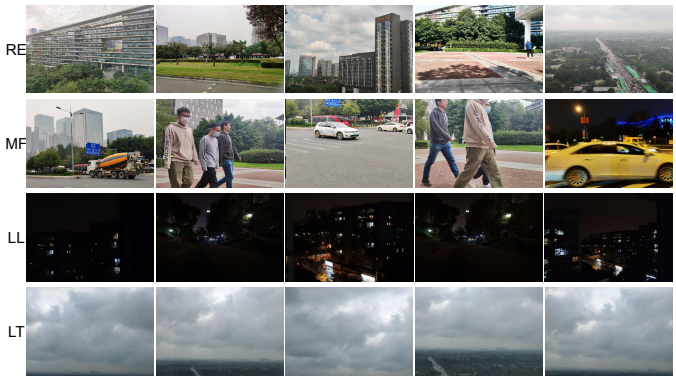


Fig. 5. A glance at our evaluation dataset. Our dataset contains 4 categories, regular(RE), moving-foreground(MF), low-light(LL) and low-texture(LT). Each category contains 350 pairs, a total of 1400 pairs, with synchronized gyroscope readings.

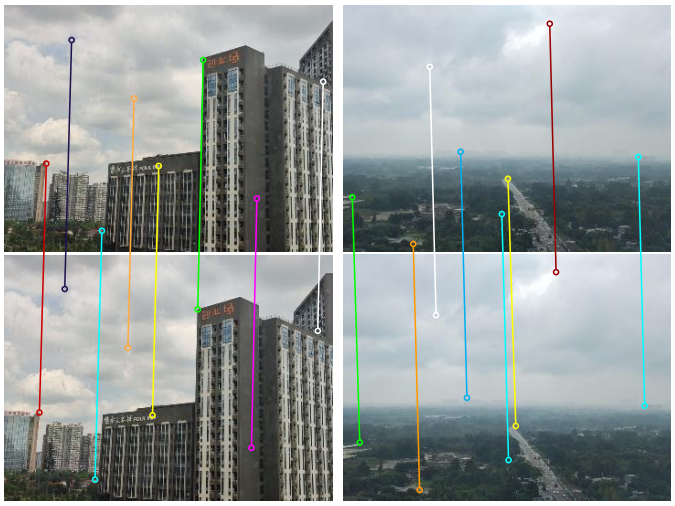


Fig. 6. We mark the correspondences manually in our evaluation set for quantitative metrics. For each pair, we mark $6 \sim 8$ point matches.

IV. Experimental Results

A. Dataset

Previously, there are some dedicated datasets that are designed to evaluate the homography estimation [14] or the image deblurring with the artificial-generated gyroscope-frame pair [36], whether none of them combine real gyroscope readings with corresponding video frames. So we propose a new dataset and benchmark GF4.

Training Set To train our network, we record a set of videos with the gyroscope readings using a hand-held OIS cellphone Xiaomi CC9pro. We choose scenes with rich textures so that sufficient feature points can be detected to calculate the Essential Mixtures model. The videos last 300 seconds, yielding 9,000 frames in total. Note that, the scene type is not important as it can provide enough features as needed for Essential Mixtures estimation.

Evaluation Set For the evaluation, we capture scenes with different types, to compare with image-based registration methods. Our dataset contains 4 categories, including regular (RE), low-texture (LT), low-light (LL), and moving-foregrounds (MF) frame-gyroscope pairs. Each scene contains 350 pairs. So, there are 1400 pairs in the dataset. We show some examples in Fig. 5. For quantitative evaluation, we manually mark $6 \sim 8$ point correspondences per pair, distributing uniformly on frames. Fig. 6 shows some examples.

TABLE III
Comparisons with non-OIS camera.

	Non-OIS Camera	OIS Camera	Ours
Geometry Distance	0.688	1.038	0.709

B. Implementation Details

Our network is trained on $9k$ rich-texture frames with resolution of 608x800 pixels over $1k$ epoches by an Adam optimizer [52] whose $l_r = 1.0 \times 10^{-4}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$. The batch size is 8, for each epoch, all 9k data are used, so there are 1125 steps for one epoch, For every 50 epochs, the learning rate is reduced by 20%. The entire training process costs about 50 hours. The implementation is in PyTorch and the network is trained on one NVIDIA RTX 2080 Ti.

C. Comparisons with non-OIS camera

1) Qualitative Comparisons: We compare our method with non-OIS cameras. In general, if the OIS motion could be compensated successfully, our method should perform equally well as non-OIS cameras. For comparison, ideally, we should use one camera with OIS turn on and off. However, the OIS cannot be turned off easily. Therefore, we use two cell phones with similar camera intrinsics, the one with OIS is Xiaomi CC9Pro equipped with Snapdragon 730G and the other one without OIS is Qualcomm QRD equipped with Snapdragon 7150, and letting them capture the same scene twice with similar motions. The evaluation set contains 4 categories, including RE, LT, LL, and MF. Each category contains 50 pairs. Fig. 7 shows some examples. Fig. 7 (a) shows the input frames. Fig. 7 (b) shows the gyro alignment on a non-OIS camera. As seen, images can be well aligned. Fig. 7 (c) shows the gyro alignment on an OIS camera. Due to the OIS interferences, images cannot be aligned directly using the gyro. Fig. 7 (d) shows our results. With OIS compensation, images can be well aligned on OIS cameras.

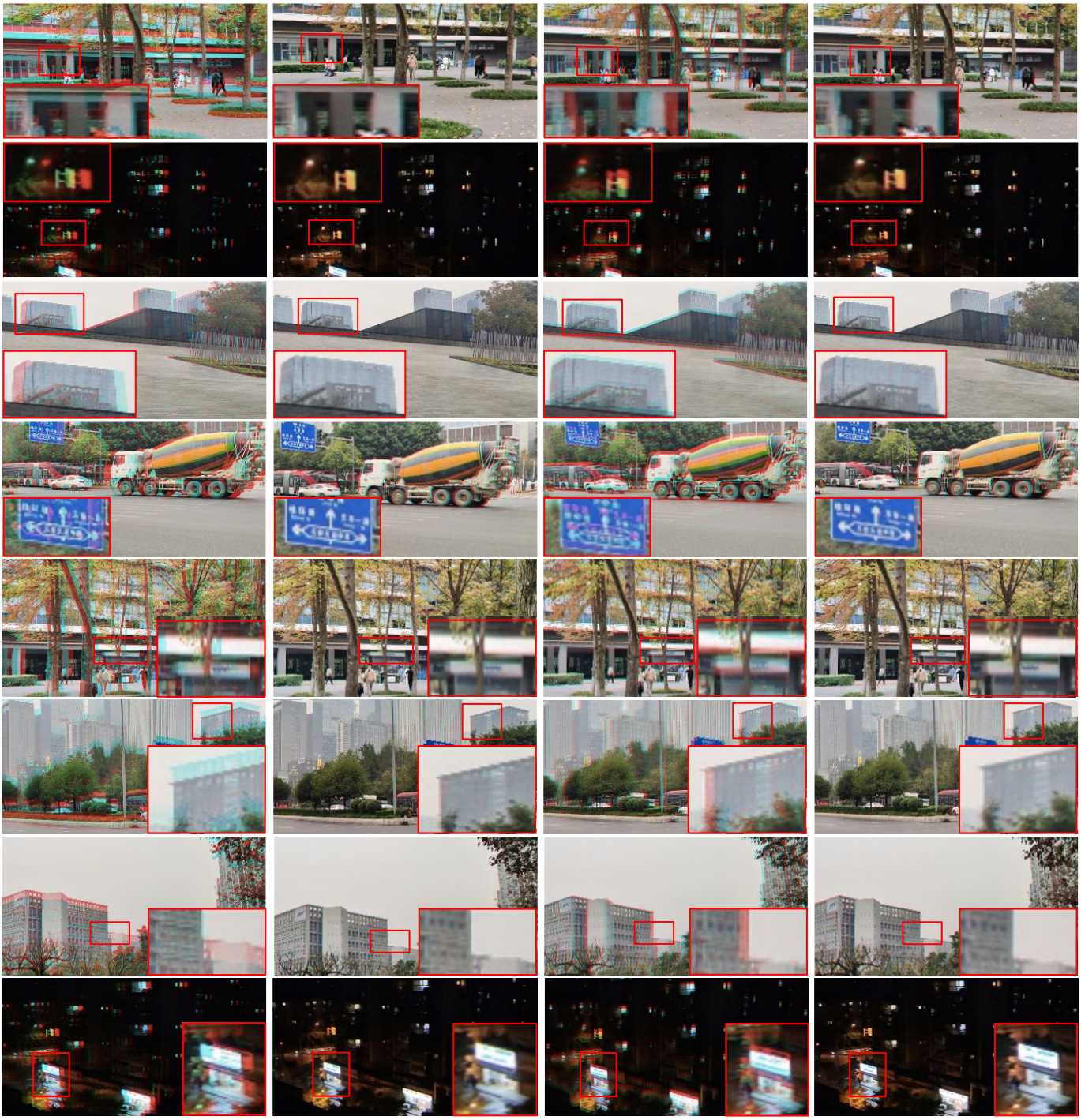
2) Quantitative Comparisons: We also calculate quantitative values. Similarly, we mark the ground-truth for evaluation. The average geometry distance between the warped points and the manually labeled GT points are computed as the error metric (the lower the better). Table III shows the results. Our result 0.709 is comparable with non-OIS camera 0.688 (slightly worse), while no compensation yields 1.038, which is much higher.

D. Comparisons with Image-based Methods

In order to show the importance and robustness of the gyro-based alignment and highlight the importance of enabling this capability on OIS cameras. We compare with image-based methods that look at image content.

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(a) Input (b) Non-OIS Camera (c) OIS Camera (d) Ours

Fig. 7. Comparisons with non-OIS cameras. (a) two input frames. (b) gyro alignment results on the non-OIS camera. (c) gyro alignment results on the OIS camera. (d) our OIS compensation results. Without OIS compensation, clear misalignment can be observed in (c) whereas our method can solve this problem and be comparable with non-OIS results in (b).

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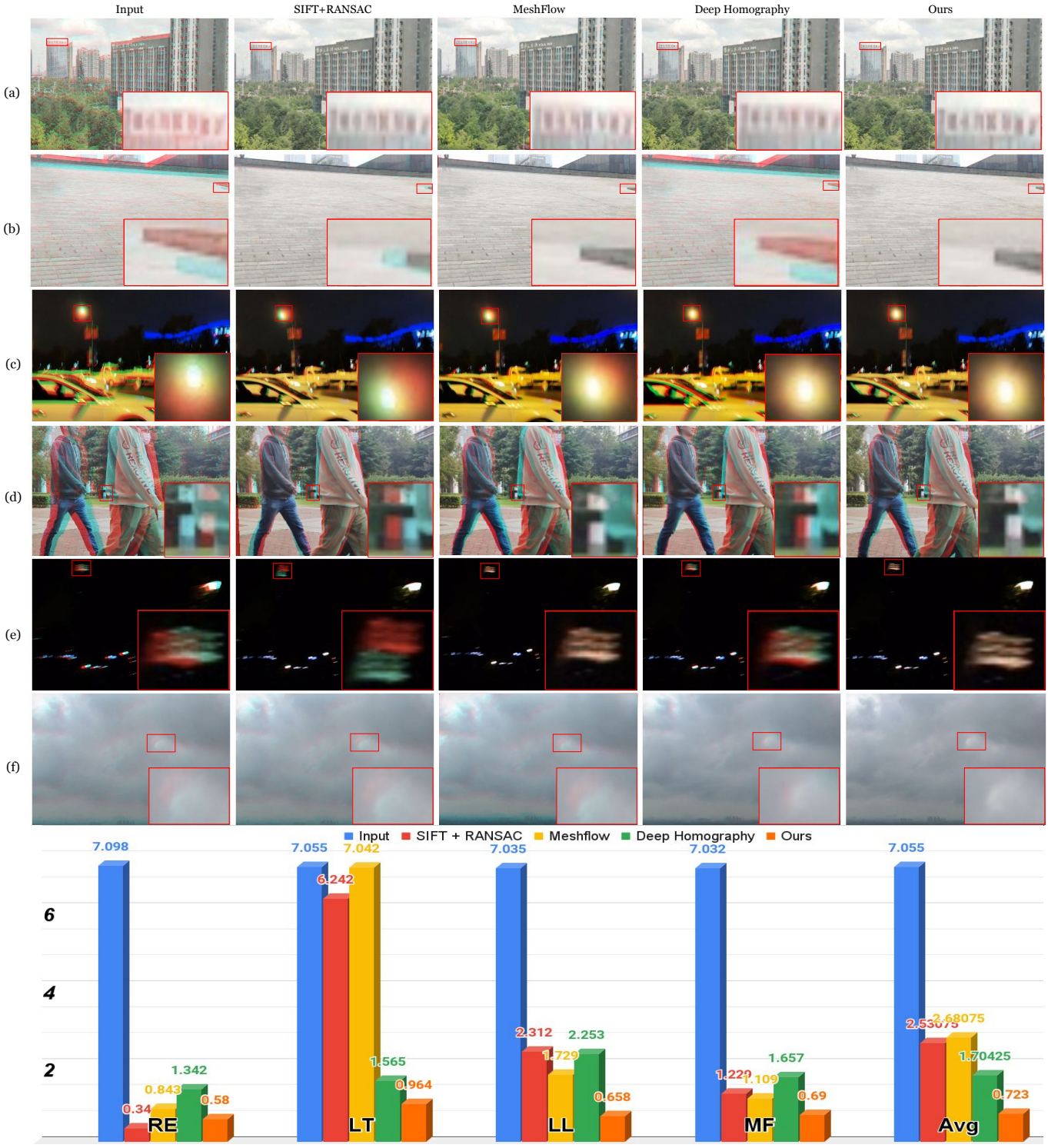


Fig. 8. Comparisons with image-based methods. We compare with SIFT [18] + RANSAC [19], Meshflow [8], and the recent deep homography [14] method. We show examples covering all scenes in our evaluation dataset. Our method can align images robustly while image-based methods contain some misaligned regions.

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TABLE IV
QUANTITATIVE COMPARISONS ON THE EVALUATION DATASET. THE BEST PERFORMANCE IS MARKED IN RED AND THE SECOND-BEST IS IN BLUE.

1)	RE	LT	LL	MF	Avg
2) $\mathcal{I}_{3\times3}$	7.098(+2785.37%)	7.055(+566.19%)	7.035(+540.13%)	7.032(+767.08%)	7.055(+639.32%)
3) GYRO + OIS	1.127(+358.13%)	1.250(+18.04%)	1.158(+5.37%)	1.295(+59.68%)	1.209(+26.54%)
4) SIFT [18]+ RANSAC [19]	0.340(+38.21%)	6.242(+489.42%)	2.312(+110.37%)	1.229(+51.54%)	2.531(+165.21%)
5) SIFT [18] + MAGSAC [53]	0.213(−13.41%)	5.707(+438.90%)	2.818(+156.41%)	0.811(+0.00%)	2.387(+150.17%)
6) ORB [54] + RANSAC [19]	0.653(+165.45%)	6.874(+549.10%)	1.136(+3.37%)	2.27(+179.90%)	2.732(+186.43%)
7) ORB [54] + MAGSAC [53]	0.919(+273.58%)	6.859(+547.69%)	1.335(+21.47%)	2.464(+203.82%)	2.894(+203.30%)
8) SOSNET [55] + RANSAC [19]	0.246(+0.00%)	5.946(+461.47%)	1.977(+79.89%)	0.907(+11.84%)	2.269(+137.78%)
9) SOSNET [55] + MAGSAC [53]	0.309(+25.61%)	5.585(+427.38%)	1.972(+79.44%)	1.142(+40.81%)	2.252(+136.00%)
10) SURF [31] + RANSAC [19]	0.343(+39.43%)	3.161(+198.49%)	2.213(+101.36%)	1.420(+75.09%)	1.784(+86.98%)
11) SURF [31] + MAGSAC [53]	0.307(+24.80%)	3.634(+243.15%)	2.246(+104.37%)	1.267(+56.23%)	1.863(+95.28%)
12) D2-Net [56] + RANSAC [19]	1.246(+406.50%)	1.419(+33.99%)	1.345(+22.38%)	2.109(+160.05%)	1.529(+60.31%)
13) D2-Net [56] + MAGSAC [53]	1.694(+588.62%)	2.005(+89.33%)	2.078(+89.08%)	2.621(+223.06%)	2.099(+119.99%)
14) R2D2 [57] + RANSAC [19]	0.596(+142.28%)	1.059(+0.00%)	1.099(+0.00%)	1.063(+31.07%)	0.954(+0.00%)
15) R2D2 [57] + MAGSAC [53]	3.483(+1315.85%)	6.441(+508.22%)	4.413(+301.55%)	2.621(+223.18%)	4.2395(+344.28%)
16) MeshFlow [8]	0.843(+242.68%)	7.042(+564.97%)	1.729(+57.32%)	1.109(+36.74%)	2.681(+180.93%)
17) Deep Homography [14]	1.342(+445.53%)	1.565(+47.78%)	2.253(+105.00%)	1.657(+104.32%)	1.704(+78.60%)
18) Ours	0.580(+135.77%)	0.964(−8.97%)	0.658(−40.13%)	0.690(−14.92%)	0.723(−24.23%)

1) Qualitative Comparisons: Firstly, we compare our method with one frequently used traditional feature-based algorithm, i.e. SIFT [18] and RANSAC [19] that compute a global homography, and another feature-based algorithm, i.e. Meshflow [8] that deforms a mesh for the non-linear motion representation. Moreover, we compare our method with the recent deep homography method [14].

Fig. 8 (a) shows a regular example where all the methods work well. Fig. 8 (b), SIFT+RANSAC fails to find a good solution, so does deep homography, while Meshflow works well. One possible reason is that a single homography cannot cover the large depth variations. Fig. 8 (c) illustrates a moving-foreground example that SIFT+RANSAC and Meshflow cannot work well, as few features are detected on the background, whereas Deep Homography and our method can align the background successfully. A similar example is shown in Fig. 8 (d), SIFT+RANSAC and Deep Homography fail. Meshflow works on this example as sufficient features are detected in the background. In contrast, our method can still align the background without any difficulty. Because we do not need the image contents for the registration. Fig. 8(e) is an example of low-light scenes, and Fig. 8(f) is a low-texture scene. All the image-based methods fail as no high-quality features can be extracted, whereas our method is robust.

2) Quantitative Comparisons: We also compare our method with other feature-based methods quantitatively via the geometry distance. For the feature descriptors, we choose SIFT [18], ORB [54], SOSNet [34], SURF [31], D2-Net [56] and R2D2 [57]. For the outlier rejection algorithms, we choose RANSAC [19] and MAGSAC [53]. The errors for each category are shown in Table IV followed by the overall averaged error, where $\mathcal{I}_{3\times3}$ refers to a 3×3 identity matrix as a reference. In particular, feature-based methods sometimes crash, when the error is larger than $\mathcal{I}_{3\times3}$ error, we set the error equal to $\mathcal{I}_{3\times3}$ error. Regarding the motion model, from 4) to 15) and 17) are single homography, 16) is mesh-based, and 3) is a homography array. In Table IV, we mark the best performance in red and the second-best in blue.

As shown, except for comparing to feature-based methods in RE scenes, our method outperforms the others for all categories. It is reasonable because, in regular(RE) scenes, a set of high-quality features is detected that allows to output a good solution. In contrast, gyroscopes can only compensate for rotational motions, which decreases scores to some extent. For the rest scenes, our method beats the others with an average error being lower than the 2nd best by 24.32%. Especially for low-light(LL) scenes, our method computes an error that is at least lower than the 2nd best by 40.13%.

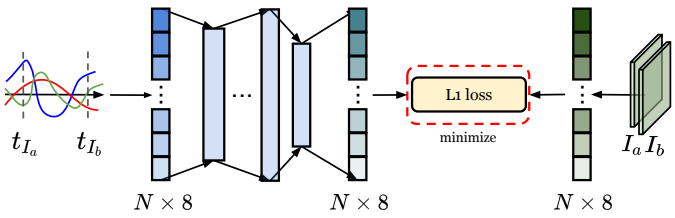


Fig. 9. Regression of the homography array using the fully connected network. For each pair of frames I_a and I_b, a homography array is computed by using gyro readings, which is fed to the network. On the other side, an Essential Mixtures model is produced as target to guide the training process.

E. Ablation Studies

1) Fully Connected Neural Network: Our network is fully convolutional, where we convert gyroscope data into homography arrays, then into flow fields which can be regarded as image input to the network. However, there is another option where we can directly input homography arrays as input. Similarly, on the other side, the Essential Mixtures are converted into rotation-only homography arrays and then used as guidance. Fig. 9 shows the pipeline, where we test two homography representations, including 3×3 homography

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matrix elements and H_{4pt} representation from [58] that represents homography by 4 motion vectors. The network is fully connected and L2 loss is adopted for the regression. We adopt the same training data as described above. The result is that both of representations do not converge, where the H_{4pt} is slightly better than directly regressing matrix elements.

Perhaps, there exist other representations or network structures that may work well or even better than our current proposal. Here, as the first try, we have proposed a working pipeline and want to leave the improvements as future works.

2) Global Essential vs. Mixtures: To verify the effectiveness of our Essential Mixtures model, we compare it with a global essential matrix. Here, we choose the evaluation dataset of the regular scenes to alleviate the feature problem. We estimate the global essential matrix and Essential Mixtures, then convert to the rotation-only homographies, respectively. Finally, we align the images with rotation-only homographies accordingly. An array of homographies from Essential Mixtures produces an error of **0.451**, which is better than the error of **0.580** produced by a single homography from a global essential matrix. It indicates that the Essential Mixtures model is functional in the case of RS cameras.

Moreover, we generate GT with the two methods and train our network, respectively. As shown in Table V, the network trained on Essential Mixtures-based GT outperforms the global essential matrix, which demonstrates the effectiveness of our Essential Mixtures.

TABLE V
THE PERFORMANCE OF NETWORKS TRAINED ON TWO DIFFERENT GT.

Ground Truth	RE	LT	LL	MF	Avg
Global Fundamental	0.930	1.189	0.769	0.917	0.951
Fundamental Mixtures	0.580	0.964	0.658	0.690	0.723

TABLE VI
THE PERFORMANCE OF NETWORKS WITH DIFFERENT BACKBONES.

	RE	LT	LL	MF	Avg
R2UNet [59]	0.716	0.988	0.649	0.711	0.766
AttUNet [60]	0.846	1.072	0.754	1.025	0.924
R2AttUNet [59], [60]	0.638	1.013	0.637	0.736	0.756
Ours	0.580	0.964	0.658	0.690	0.723

TABLE VII
THE INFERENCE TIME OF NETWORKS WITH DIFFERENT BACKBONES.

	R2UNet [59]	AttUNet [60]	R2AttUNet [59], [60]	Ours
Time	122.930 ms	50.333 ms	127.899 ms	30.084 ms

3) Backbone: We choose the UNet [49] as our network backbone, we also test several other variants [59], [60]. Except for AttUNet [60], performances are similar, as shown in Table VIII. Note that R2AttUNet [59], [60] is the combination of R2Unet and AttUnet which is implemented by ourselves. Moreover, the computing complexity comparisons are illustrated in Fig. VII at an input resolution of $(512, 640)$, we find that achieving similar performances, UNet [49] is 4 times faster than R2AttUNet.

4) Loss function: L1loss is adopted for the regression in our experiments. We also test several strong regression loss functions including HingeLoss, XSigmoid, L2loss, LogCosh, and SmoothL1. The model trained by L1loss performs the best on average, and HingeLoss is more suitable to solve classification problems, so it performs poorly on our regression problems.

TABLE VIII
THE PERFORMANCE OF NETWORKS WITH DIFFERENT LOSS FUNCTIONS.

	RE	LT	LL	MF	Avg
HingeLoss	7.202	7.410	7.058	7.044	7.179
XSigmoid	0.598	0.969	0.665	0.744	0.744
L2loss	0.573	0.967	0.670	0.753	0.741
LogCosh	0.580	0.957	0.649	0.746	0.733
SmoothL1	0.577	0.954	0.688	0.697	0.729
Ours	0.580	0.964	0.658	0.690	0.723

V. CONCLUSION

We have presented a DeepOIS pipeline for the compensation of OIS motions for gyroscope image registration. We have captured the training data as video frames as well as their gyro readings by an OIS camera and then calculated the ground-truth motions with our proposed Essential Mixtures model under the setting of rolling shutter cameras. For the evaluation, we have manually marked point correspondences on our captured dataset for quantitative metrics. The results show that our compensation network works well compared with non-OIS cameras and outperforms other image-based methods. In summary, a new problem is proposed and we show that it is solvable by learning the OIS motions, such that gyroscope can be used for image registration on OIS cameras.

The main limitation of our framework is that it is restricted to rotations only. For some applications, such as video stabilization, combining gyroscope data and image content could improve the accuracy and robustness of constructing camera motion. Future work could investigate fusing sensor data and image content to compensate for the OIS motion.

Our code and dataset will be released to facilitate more research in this direction.

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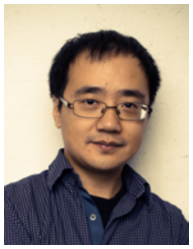
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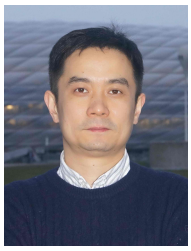
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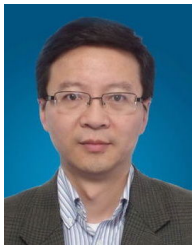
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